



Question one (10 marks): choose the answer that best describes the following sentences.

(50's 1)

- Electron beam directed only to the points of picture to be displayed.
 - Raster Scan
 - Random Scan
 - Interlaced Scanning
 - Cathode ray tube
- A part of memory for graphics that has the intensity value for all the screen coordinates.
 - Frame buffer
 - Pixels
 - Display controller
 - Cathode ray tube
- Algorithms that identify those portions of a picture either inside or outside of a specified region of space.
 - Rendering
 - Filling
 - Cropping
 - Clipping
- A _____ is a chain of quadratic Bezier curves joined end-to-end.
 - spline
 - cubic Bezier splines
 - quadratic Bezier splines
 - cubic Bezier curves
- A display device in which polarized light is passed through a liquid crystal material.
 - CRT
 - LED
 - LCD
 - Plasma
- A Cubic Bezier curve is just a blend of:
 - two linear Bezier curves
 - two quadratic Bezier curves
 - three quadratic Bezier curves
 - four linear Bezier curves
- Which of the following is an example for Hardware-oriented color models?
 - CMYK
 - HLS
 - CIE Lab
 - HSV
- Produces jagged lines that are plotted as discrete points.
 - Cathode ray tube
 - Random Scan
 - Interlaced Scanning
 - Raster Scan
- Quadratic Bezier curves have _____ endpoints and _____ guide point.
 - one, two
 - two, one
 - two, two
 - three, two
- A two-dimension transformation that have matrix $M = \begin{bmatrix} 1 & h & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.
 - Scaling in X direction
 - Scaling in Y direction
 - Shear in X direction
 - Shear in Y direction

Question two (10 marks): answer the following questions.

(SO's: 1, 6)

1. Derive the equation of the decision parameter p_k for the k^{th} step in the Bresenham's line algorithm for the slope $|m| < 1$. (your answer should include p_{k+1} and p_0)
2. Calculate the required points to plot the circle with radius = 6 using the midpoint circle algorithm in the first quadrant.
3. Given an ellipse with parameters $r_x = 4$ and $r_y = 3$ and center (0, 0), trace the midpoint ellipse algorithm in the first quadrant.

Hints: $p_{10} = r_y^2 - r_x^2 r_r + \frac{1}{4} r_r^2$ If $(p_{1k} < 0)$ $p_{1k+1} = p_{1k} + 2r_y^2 x_{k+1} + r_y^2$ Otherwise $p_{1k+1} = p_{1k} + 2r_y^2 x_{k+1} + r_y^2 - 2r_x^2 y_{k+1}$	$p_{20} = r_y^2 \left(x_0 + \frac{1}{2} \right)^2 + r_x^2 (y_0 - 1)^2 - r_x^2 r_y^2$ If $(p_{2k} < 0)$ $p_{2k+1} = p_{2k} - 2r_x^2 y_{k+1} + r_x^2$ Otherwise $p_{2k+1} = p_{2k} - 2r_x^2 y_{k+1} + r_x^2 + 2r_y^2 x_{k+1}$
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Question three (10 marks): answer the following questions.

(SO's: 1, 6)

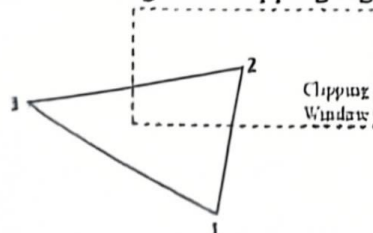
1. Illustrate with drawing the two-dimensional viewing-transformation pipeline.
2. Given a graphics package that provides only a rotate function with respect to the coordinate origin:
 - a. Clarify with drawing how to generate a 2D rotation about any pivot point (x_r, y_r) .
 - b. Derive a general formula for this rotation in (a.).
 - c. Apply the general formula, you derived in (b.), to rotate a triangle with vertices (3, 1), (4, 1) and (3, 4) by 90° about its pivot point (3, 2).
3. Given a window in the world coordinates $(W_l, W_r, W_b, W_t) = (2, 10, 30, 40)$ and a viewing port with coordinates $(V_l, V_r, V_b, V_t) = (50, 100, 250, 300)$, how would you map a point at (8, 35) in the world coordinates to the view port.

Question Four (10 Marks): Answer the following questions.

(SO's: 1, 6)

1. A Bézier curve is a parametric curve in which a set of control points defines a smooth, continuous curve by means of a formula.
 - a. Drive the quadratic Bézier curves formula.
 - b. What is the quartic Bézier curves formula.
2. Clip a line with endpoints A(8, -2) and B(28, 14) using the Cohen Sutherland algorithm with the following clipping window:

$$xw_{\min} = 6, \quad yw_{\min} = 4, \quad xw_{\max} = 21, \quad yw_{\max} = 11.$$
3. Given the following set of polygon vertices {1, 2, 3} and a clipping window, illustrate how to clip the polygon edges using the Sutherland-Hodgeman clipping algorithm.



4. The scan-line filling algorithm is performed by first determining the intersection positions of the boundaries of the fill region with the screen scan lines. Illustrate using drawing how to compute x intercepts incrementally using integer operations for an edge with endpoints (2,3) and (5,10).

- Good Luck -